

ZHANGWU, China

WMO: 542360

Lat: 42.3842N

Lon: 122.5492E

Elev: 84

StdP: 100.32

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-19.3	-30.3	0.2	-15.1	-28.2	0.3	-14.7	11.2	-3.2	9.7	-4.9	1.8	320	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	31.9	21.9	30.6	22.0	29.4	21.6	25.6	28.9	24.8	27.9	24.1	27.0	4.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	19.8	27.7	23.8	18.9	26.7	23.1	18.0	26.1	79.5	29.0	76.2	28.0	73.2	27.1	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	9.7	8.3	-24.7	34.3	3.0	1.8	-26.8	35.6	-28.6	36.6	-30.3	37.6	-32.5	38.9
			-25.0	26.9	2.9	0.9	-27.1	27.6	-28.8	28.1	-30.5	28.6	-32.6	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-11.1	-6.2	1.2	10.2	17.5	22.2	24.7	23.5	17.9	9.8	-0.5	-8.7
	DBStd	13.14	4.31	4.87	4.70	4.19	3.46	2.61	2.00	2.49	3.36	4.56	5.57	4.44
	HDD10.0	2387	653	454	273	48	1	0	0	1	61	316	581	
	HDD18.3	4152	911	687	530	246	58	3	0	1	46	267	564	839
	CDD10.0	1817	0	0	1	53	232	365	456	417	238	53	2	0
	CDD18.3	541	0	0	0	1	31	118	197	160	33	1	0	0
	CDH23.3	4352	0	0	0	16	336	956	1564	1194	278	7	0	0
	CDH26.7	1244	0	0	0	1	78	318	475	328	43	1	0	0
Wind	WSAvg	3.6	3.0	3.6	4.3	4.9	4.5	3.8	3.4	2.9	3.0	3.5	3.4	2.9
Precipitation	PrecAvg	512	2	2	8	26	41	79	150	107	62	25	7	3
	PrecMax	744	9	15	60	114	104	164	332	254	215	94	26	20
	PrecMin	330	0	0	0	2	14	19	41	16	4	0	0	0
	PrecStd	107	3	4	12	21	25	38	73	59	42	23	7	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.5	9.3	17.6	25.4	31.3	34.0	33.6	32.6	29.6	24.3	16.1	6.1
		MCWB	-0.8	2.3	8.3	12.9	16.9	19.5	23.6	23.7	20.2	16.2	10.9	3.8
	2%	DB	0.3	6.3	14.1	22.8	28.7	31.8	31.9	30.8	27.9	21.6	13.0	3.0
		MCWB	-2.9	1.0	6.2	12.1	16.2	19.3	23.6	23.7	19.2	14.4	9.0	0.0
	5%	DB	-1.7	3.9	11.5	20.3	27.0	30.0	30.5	29.7	26.3	19.5	10.6	0.7
		MCWB	-4.5	-0.2	4.6	10.7	15.1	19.2	23.3	23.2	18.2	13.3	6.8	-1.7
	10%	DB	-3.7	1.7	9.2	18.1	25.1	28.4	29.3	28.6	24.9	17.6	8.3	-1.3
		MCWB	-6.1	-1.8	3.1	9.4	14.2	19.0	23.0	22.6	17.5	11.9	5.0	-3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	3.6	9.5	15.6	19.6	23.5	27.0	26.7	23.3	18.4	12.3	4.6
		MCDB	1.7	7.1	15.1	21.5	27.1	28.8	31.7	30.2	26.8	22.2	14.8	5.6
	2%	WB	-2.5	1.7	7.1	13.6	18.2	22.2	25.8	25.6	21.5	16.5	9.7	0.8
		MCDB	-0.3	5.4	12.6	20.0	25.3	27.4	29.4	28.8	25.4	19.6	12.2	2.3
	5%	WB	-4.3	-0.1	5.3	12.0	17.0	21.5	25.0	24.8	20.4	15.1	7.5	-1.6
		MCDB	-1.9	3.4	10.5	18.1	23.4	26.3	28.2	27.9	23.9	18.0	9.6	0.6
	10%	WB	-6.1	-1.6	3.6	10.6	15.8	20.7	24.4	24.1	19.3	13.4	5.1	-3.8
		MCDB	-3.7	1.6	8.5	16.7	22.3	25.2	27.4	27.0	22.7	16.4	8.1	-1.5
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.0	10.8	11.7	11.7	10.0	8.1	8.9	12.2	11.7	10.0	10.2
		MCDBR	12.9	13.2	14.6	15.3	15.3	13.7	10.2	10.2	13.7	13.8	12.5	12.1
		MCWBR	10.3	9.2	8.6	7.5	6.0	4.6	4.1	4.3	6.2	7.5	8.8	10.0
	5% WB	MCDBR	12.6	12.2	13.1	12.5	12.5	9.8	7.6	7.8	10.3	11.1	10.6	11.8
		MCWBR	10.5	9.1	8.4	6.9	5.9	4.2	4.0	4.1	5.8	7.4	8.7	10.2
Clear-Sky Solar Irradiance	taub	0.334	0.392	0.451	0.556	0.624	0.704	0.681	0.597	0.522	0.508	0.411	0.350	
	taud	2.103	1.967	1.875	1.693	1.613	1.541	1.634	1.786	1.879	1.817	1.978	2.072	
	Ebn at Noon	770	775	776	724	688	634	644	685	703	639	660	705	
	Edh at Noon	105	143	178	230	256	276	249	207	176	165	120	100	
All-Sky Solar Radiation	RadAvg	2.39	3.31	4.47	5.34	5.98	5.47	5.16	5.02	4.55	3.37	2.35	1.99	
	RadStd	0.19	0.22	0.27	0.36	0.45	0.46	0.47	0.33	0.21	0.22	0.19	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	-0.91	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page